

Macro Three Replication Project: Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

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1 The CFA Franc Zone

The CFA franc zone is one of the longest-running fixed exchange rate arrangements in the global economy. It links a group of West and Central African countries to an external anchor currency through two monetary unions: the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CEMAC). The institutional architecture has historically rested on four central features: a fixed parity against the anchor currency, convertibility guarantee by the French Treasury, free capital mobility, and the pooling of foreign exchange reserves. These features have made the CFA franc zone an important case study in international macroeconomics because it combines monetary credibility and exchange-rate stability with a substantial loss of nominal exchange rate flexibility and monetary policy autonomy. The central policy question is whether this arrangement has supported or constrained economic performance.

2 Replication of the Original Study

In their article, Alassane Diallo and Adama Ba sought to explore the impact of the peg from the French franc to the euro on the economies of the countries in the CFA-zone. There are a lot of conflicts in the literature about whether they have experienced a stronger or weaker growth after the peg shift, compared to what would have been without France joining the euro-zone. On one hand we could expect a high growth rate due to the access to an intensive trade market with the European Economic Community (Dearden, 1999; Hadjimichael and Galy, 1997). But on the other hand there are many uncertainties linked to the consequence of being pegged to a strong currency like the euro, this could undermine expectations following the switch of anchor currencies (Zafar, 2005). At the time of the peg switch, in 2002, a report of the French Senate stated that there was no comprehensive study that would prove that the peg of the CFA-Franc to the euro would be an advantage or a drawback for member countries. Therefore, the authors of the paper wanted to address this gap by estimating the impact of the euro changeover on CFA-zone economies. Their goal was first to allow for more empirical evidence to strengthen the debates but also to inform CFA-zone policy makers. For their analysis they chose to use GDP per capita as the outcome variable of interest.

Diallo and Ba (2024) use the synthetic control method (Abadie, 2021) to construct a proxy GDP per capita for each CFA-countries, meaning an estimate of what the growth would have been without the peg changeover. This is done through the construction of a weighted average of comparable non-CFA countries, based on pre-treatment characteristics. This method provides a transparent comparison group for our treated countries, and it can produce a more credible counterfactual than with a single control country.

The thirteen treated countries of the CFA-zone are divided between two economical communities: the West African Economic and Monetary Community (WAEMU) and the Central African Economic and Monetary Community (CAEMC), as shown in Table 1.

The donor countries all have fixed exchange rates regimes and have been matched based on other economic or institutional characteristics. The authors first included African countries with similar socioeconomic characteristics as the treated countries (Botswana, Eswatini, Lesotho, Mauritius, Morocco, Namibia and Seychelles) and they then added other countries with the similar level of development (Bangladesh, Barbados, Bhutan, Bolivia, Dominican Republic, Ecuador, Grenade, Laos, Oman, Panama, Saint Kitts & Nevis and Saint-Lucia). We have chosen to add three countries that are also similar to allow for more precise results (Turkmenistan, Guyana and Iraq).

Table 1: Treated countries and regional classification

Country	ISO3	Zone	Entry
Benin	BEN	WAEMU	Original CFA member
Burkina Faso	BFA	WAEMU	Original CFA member
Côte d’Ivoire	CIV	WAEMU	Original CFA member
Mali	MLI	WAEMU	Rejoined WAEMU in 1984
Niger	NER	WAEMU	Original CFA member
Senegal	SEN	WAEMU	Original CFA member
Togo	TGO	WAEMU	Original CFA member
Guinea-Bissau	GNB	WAEMU	Joined WAEMU in 1997
Cameroon	CMR	CEMAC	Original CFA member
Central African Republic	CAF	CEMAC	Original CFA member
Chad	TCD	CEMAC	Original CFA member
Republic of Congo	COG	CEMAC	Original CFA member
Gabon	GAB	CEMAC	Original CFA member
Equatorial Guinea	GNQ	CEMAC	Joined CEMAC in 1985

The treatment date is 2002, as it was the date of the French Franc to Euro peg changeover. This means that the synthetic control weights will be computed on the similarities of the characteristics from 1980 to 2002.

The paper finds positive post-treatment output gaps for several countries, however those results are not statistically significant for any countries, except for Equatorial Guinea.

3 Replication Methodology and Data

The replication follows the original paper’s country-level synthetic control design. For each treated CFA franc country, we estimate a counterfactual GDP path using a weighted average of non-CFA donor countries, and then compare the observed post-2002 GDP path with this synthetic counterfactual. The treatment year, time period, and country sample all follow the original paper’s specifications, while variable selection and construction, and missing data approaches diverged. Further details on the variable construction, and missing-data treatment are reported in Appendix A: Table 3, and Table 4.

Let Y_{it} denote GDP per capita for country i in year t . For a treated country $i = 1$, the unobserved counterfactual outcome after treatment is approximated using a weighted average of J donor countries:

$$\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j Y_{jt}, \quad (1)$$

where the donor weights satisfy the standard synthetic control constraints:

$$w_j \geq 0, \quad \sum_{j=2}^{J+1} w_j = 1. \quad (2)$$

The weights are chosen to make the synthetic control resemble the treated country as closely as possible before treatment. Let X_1 denote the vector of pre-treatment characteristics for the treated country, and let X_0 denote the corresponding matrix for the donor countries. The donor-weight vector W is selected to minimise the distance between the treated country and the weighted donor average:

$$\min_W (X_1 - X_0 W)' V (X_1 - X_0 W), \quad (3)$$

subject to the constraints in Equation 2. The matrix V determines the relative importance assigned to each predictor in matching the treated country to the donor pool. In the reconstructed specification, the predictors

include agriculture, government share, investment share, official development assistance, foreign direct investment, labour, institutional quality, and selected pre-treatment GDP values. Industry is excluded from the final specification because of missingness and data-quality concerns, as discussed in Appendix A.

The estimated treatment effect is the post-treatment gap between the observed outcome and the synthetic counterfactual:

$$\hat{\alpha}_{1t} = Y_{1t} - \hat{Y}_{1t}^N = Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt}, \quad t \geq 2002. \quad (4)$$

A positive value of $\hat{\alpha}_{1t}$ indicates that actual GDP per capita was above the synthetic counterfactual, while a negative value indicates that actual GDP per capita was below the counterfactual. The interpretation of these gaps depends on the quality of the pre-treatment fit: if the synthetic control does not reproduce the treated country’s pre-2002 path closely, the post-treatment gap is less informative.

The replication is best understood as a reconstructed replication rather than a mechanical reproduction of the original paper. Two issues were especially important. First, the GDP series used in the original paper could not be identified exactly from the published documentation. Second, the paper does not report how missing observations were handled, even though missing outcomes or predictors can affect donor eligibility, predictor averages, and estimated synthetic controls. These issues are discussed in Appendix A; the final variable definitions are summarised in Table 3, and the missing-data procedures are summarised in Table 4.

The analysis proceeds in four steps:

Step	Description
1	The country-year panel is reconstructed from public datasets.
2	Variables are harmonised and missing values are handled using documented rules.
3	One synthetic control is estimated for each treated country, producing donor weights, predictor-balance statistics, actual-versus-synthetic GDP paths, and post-treatment gaps.
4	In-space placebo tests are used to compare the treated-country gaps with gaps obtained when the same procedure is applied to donor countries as placebo-treated units.

Several additional reporting issues in the original paper also shaped the replication strategy. These include donor-pool ambiguity, inconsistent country naming, unclear variable definitions, inflation-source inconsistencies, table-label errors, and numerical values that are difficult to reconcile with the stated data structure. These issues do not invalidate the paper’s research question, but they reduce confidence that the published tables provide sufficient information for an exact replication. The purpose of the replication is therefore to evaluate the robustness of the paper’s broad conclusions using a transparent, reproducible data construction and estimation procedure.

4 Replication Results

The country weights computed by the synthetic control method, which then allows us to create the synthetic GDP per capita, are shown in Table 4. We were not able to replicate the original values exactly as the GDP data were not the same. However, as in the original paper, the synthetic control analysis relies heavily on Bangladesh and Laos. The first has a positive weight for 9 out of the 13 treated countries, being especially present for Mali as well as Burkina Faso. Laos has positive weights for 10 out of the 13 countries, an explanation for this is that the country has similar pre-treatment income to a lot of the treated countries compared to the other donor countries. Iraq, one of the countries that we added, had positive weights for 9 out of 13 treated countries.

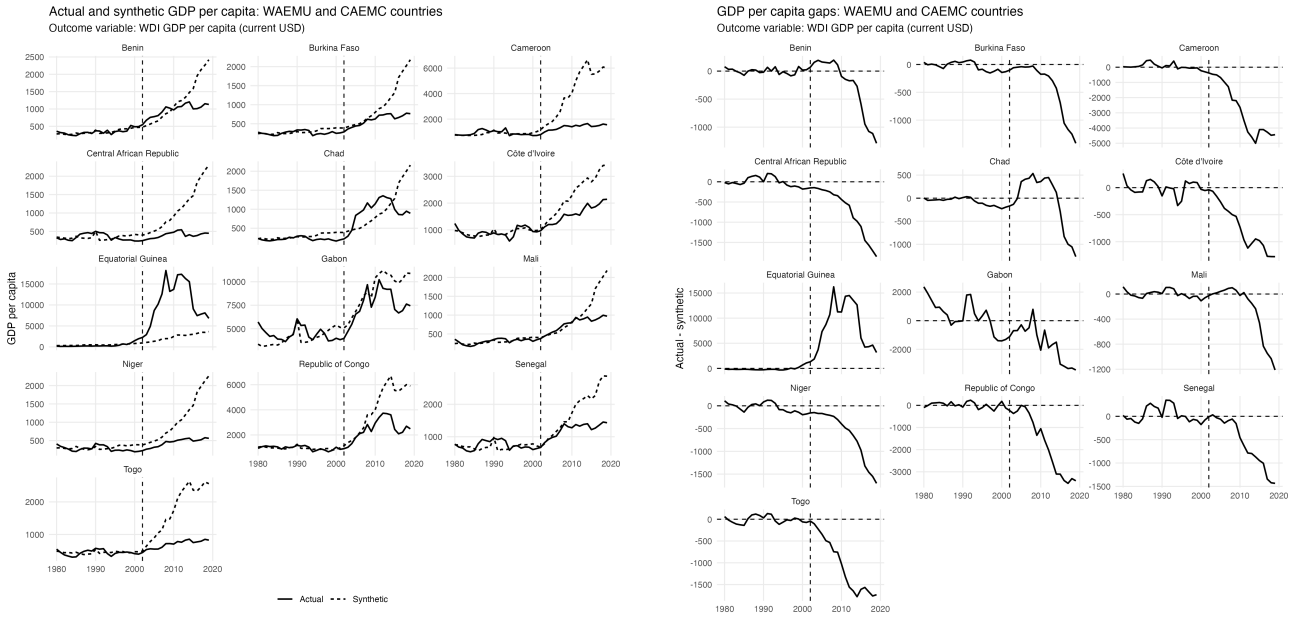
Figure 1a presents the result of our analysis using the synthetic control method, with the comparison of the synthetic control GDP (dotted line) per capita and the actual GDP per capita (solid line) for each treated country. The vertical dotted line marks the year 2002, our treatment, and thus before it both curves should be similar to attest to the good fit of our method. Thus the closer the dashed line is to the full line during the 1980 to 2002 pre-treatment period, the better the synthetic control matches the treated country. And even

Table 2: Estimated synthetic control weights by treated country

Donor country	Benin	B. Faso	Côte d'Ivoire	Mali	Niger	Senegal	Togo	Cameroon	CAR	Chad	RC	Gabon	Eq. G.
Bangladesh	.771	.901	.353		.946	.757	.356	0	.072	.785	.934	0	0
Barbados	.002	0	0		0	0	0	0	0	0	0	.345	0
Bhutan	.124	0	0		0	0	0	0	.053	0	0	0	.988
Bolivia	0	0	.093		0	0	.261	.011	.002	0	0	.075	0
Botswana	0	0	0		0	0	0	0	0	0	.007	0	0
Cabo Verde	0	0	0		0	0	0	0	0	0	0	0	0
Dominica	0	0	0		0	0	0	0	0	0	0	0	.012
Ecuador	.001	0	.110		0	0	0	0	.013	0	0	.159	0
Eswatini	0	0	0		.008	0	0	0	0	0	0	0	0
Grenada	0	0	0		0	0	0	0	0	0	0	0	0
Lao PDR	.094	.097	.120		.040	.236	.047	.301	.017	.191	.066	0	0
Lesotho	0	0	0		0	0	.203	.486	0	0	0	0	0
Mauritius	0	0	0		0	0	0	.083	0	0	0	0	0
Morocco	0	0	.225		0	0	0	0	0	0	0	0	0
Namibia	0	0	.074		0	0	.090	0	0	0	0	.407	0
St. Kitts and Nevis	0	0	0		0	0	0	0	0	0	0	0	0
St. Lucia	0	0	0		0	0	0	0	0	0	0	0	0
Seychelles	0	0	0		0	0	0	0	0	0	0	0	0
Turkmenistan	0	0	0		0	0	0	.187	.738	0	0	.725	0
Guyana	0	0	0		0	0	0	0	0	0	0	0	0
Iraq	.008	.002	.026		.006	.008	.043	.015	.022	.023	0	.033	.249

though we can note some imperfection, especially when the growth is highly variable, the synthetic and treated pathways are very similar for most countries.

Figure 1b presents the difference between the synthetic and actual GDP per capita. We can note again the good fit of our analysis as in the 1980-2002 period the gaps are generally near 0. This suggests that we have credible counterfactuals that allow us to address the euro-peg period.



(a) Actual and synthetic GDP per capita for treated CFA countries

(b) GDP per capita gaps between actual and synthetic controls for treated CFA countries

Figure 1: Synthetic control method outputs

Both figures are tools to compare the growth that actually happened with the treatment and our proxy growth. In our result, we found that almost all countries showed increasingly large negative GDP per capita gaps as the actual output fell below our counterfactual, with only Equatorial Guinea having a positive output gap. This suggests that the changeover to a euro peg had negative effects on the countries of the CFA-zone. Our results can be explained by an overvaluation of their currency because of the peg to a strong euro. This would make trade more difficult for our treated countries. However we cannot draw those conclusions using only the GDP per capita as it has multiple different input, it would thus be interesting to overview another output that is more closely related to trade, for example imports and exports.

However our results conflict with the one of the original paper. Indeed, Diallo & Ba (2024) found a positive output gap for multiple countries, for instance Chad, the Central African Republic or Mali. Thus their result

would suggest that the peg of CFA Franc to the euro may boost the real GDP per capita compared to the synthetic output. Due to the data and methodology issues in the original paper, we cannot be sure about the exact reasons for this discrepancy, but it is likely that the different GDP series, missing-data treatments, or errors in the original implementation played a major role.

4.1 Placebo testing

Figure 2 presents the results of our placebo significance testing. We followed the in-place placebo method, which consists of analysing each donor country as if it had received the treatment. Then we have estimated the synthetic control for each donor country and we can then compute the output gap for each country. The aim of this approach is to compare the difference between the pre-treatment and post-treatment output gap of each treated country with the analogous output gaps of the donor countries.

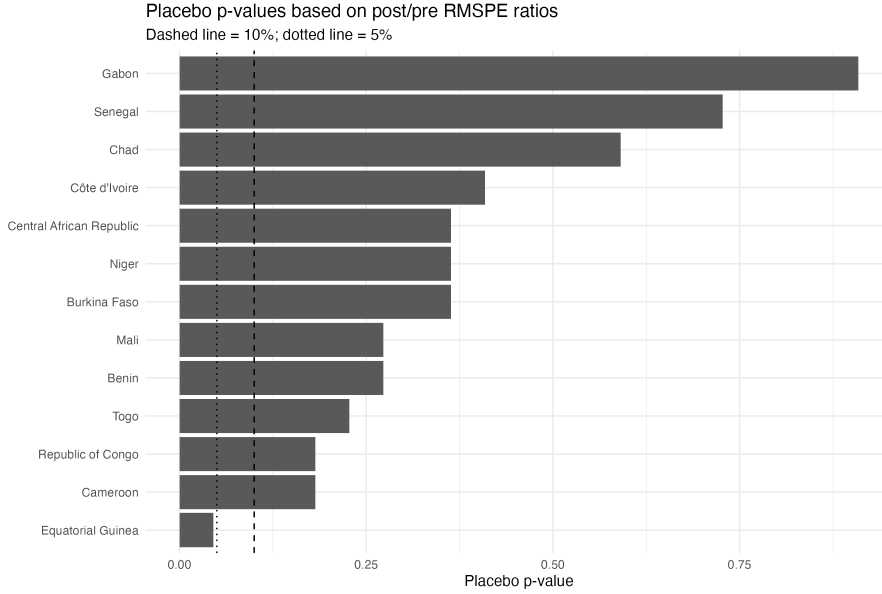


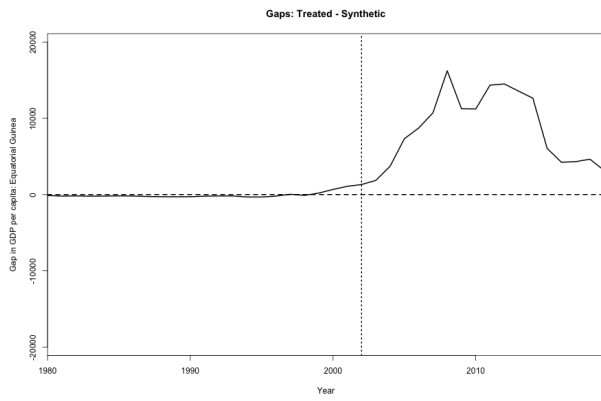
Figure 2: Placebo p-values for the synthetic control method

The p-values are computed using the post/pre Root Mean Square Per Error ratios. The ratio is computed for all countries, and we then rank the results of the donor countries. Then we can observe how unusual, relatively, the ratio of each country is by seeing how low or high in the ranking it is, with the share of countries with a more extreme ratio determining the placebo p-values.

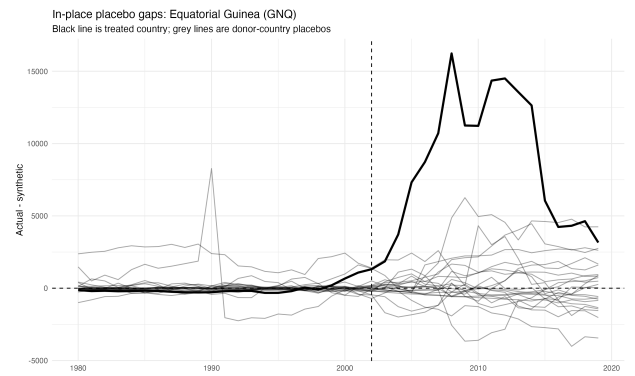
With this method, we obtained the same results as the authors of the original paper. At the 5 and 10% level, we find that only the result of Equatorial Guinea was statistically significant, while all the other treated countries do not have any significant results.

4.1.1 Equatorial Guinea

Equatorial Guinea is an unusual case. In Figure 3a we can observe the result for its synthetic control, with the output gap. And indeed we observe a consistently positive GDP per capita gap. We can also note that those results are the one that match the most those of the original paper, in terms of closeness of the numbers of the output gap. In Figure 3b, we can observe graphically that those results are significant. This suggests that the changeover to the euro peg had a significantly positive impact on Equatorial Guinea. However it is important to highlight that Equatorial Guinea is a very singular country. Indeed the authors of the original paper reference studies that show an undervaluation of its exchange rates (Feindouno et al., 2020). We can also acknowledge that there was a large discovery of oil in Equatorial Guinea's land. Since then it has become one of Sub-Saharan Africa's largest oil producers. Because of the closeness of this event and the changeover to the euro peg, we cannot conclude on if the changeover had an impact on the growth of the country. The economic boom can be easily explained by their development on the oil market, thus bringing a significant bias in our analysis.



(a) Equatorial Guinea's GDP per capita gap between actual and synthetic control



(b) Placebo gaps for Equatorial Guinea compared to donor countries

Figure 3: Results for Equatorial Guinea

A Data Construction and Variable Definitions

Table 3: Variable construction in the original paper, updated replication, and trade extension

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Real GDP per capita / selected GDP per capita	<code>gdp_selected;</code> <code>wdi_gdp_pc_current</code>	Main GDP outcome in replication	World Development Indicators	The original paper studies real GDP per capita. In the replication, alternative GDP measures were compared and WDI GDP per capita, current US dollars, was selected as the closest empirical match to the paper's reported results.	Main divergence from the original paper; The exact GDP series in the paper could not be reproduced unambiguously, considering numerous WDI and PWT indicators.
Agriculture value added	<code>agriculture</code>	SCM predictor	WDI and FAOSTAT	Agriculture, forestry, and fishing, value added as a percentage of GDP. WDI indicator: <code>NV.AGR.TOTL.ZS</code> . Alternative agriculture series were used to fill major gaps.	Retained in the final predictor set.
Industry value added	<code>industry</code>	Original/intended SCM predictor	WDI	Industry including construction, value added as a percentage of GDP. Main WDI indicator: <code>NV.IND.TOTL.ZS</code> .	Included in the original paper's predictor set but removed from the final replication because of missingness and data-quality problems.
Government expenditure share	<code>govt_share</code>	SCM predictor	Penn World Table	Government consumption share from PWT, using <code>csh_g</code> .	Corresponds to the government expenditure control described in the original paper.
Investment share / gross fixed capital formation	<code>invest_share</code>	SCM predictor	Penn World Table	Investment or capital-formation share from PWT, using <code>csh.i</code> .	Corresponds to the paper's gross fixed capital formation control.
Official development assistance	<code>oda_share</code>	SCM predictor	World Development Indicators	Main WDI series used include net ODA received. <code>DT.ODA.ODAT.GD.ZS</code> .	ODA as a share of national income. Retained in the final predictor set.
Foreign direct investment	<code>fdi</code>	SCM predictor	World Development Indicators	Foreign direct investment, net inflows as a percentage of GDP. WDI indicator: <code>BX.KLT.DINV.WD.GD.ZS</code> .	Retained in the final predictor set.

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Labour / employment	labour	SCM predictor	Penn World Table	Employment as a share of population, constructed as $100 * \text{emp} / \text{pop}$.	Corresponds to the original paper's labour factor, described as the proportion of the population in work.
Political regime / institutions	polity2	SCM predictor	Polity5	Polity2 institutional score, ranging from autocracy to democracy (-10 to +10).	Interpolation and extrapolation were applied where needed to obtain usable pre-treatment panels.
1994 CFA devaluation dummy	devaluation_1994	Original control / historical adjustment	Constructed	Dummy variable intended to capture the January 1994 CFA franc devaluation.	The original paper mentions this control. In the replication code, it was not retained as a standard SCM predictor because it has no useful cross-donor variation.
Trade openness	trade_openness	Trade-extension outcome	World Development Indicators	Exports plus imports of goods and services as a percentage of GDP. WDI indicator: NE.TRD.GNFS.ZS .	Used to test whether the euro peg changed aggregate trade exposure.
Exports of goods and services	exports_gdp	Trade-extension outcome	World Development Indicators	Exports of goods and services as a percentage of GDP. WDI indicator: NE.EXP.GNFS.ZS .	Used to examine export intensity.
Imports of goods and services	imports_gdp	Trade-extension outcome	World Development Indicators	Imports of goods and services as a percentage of GDP. WDI indicator: NE.IMP.GNFS.ZS .	Used to examine import intensity.
Exports to EU share	XEU	Trade-extension outcome	IMF Direction of Trade Statistics	Computed as 100 times exports to the EU divided by exports to the world.	Measures export composition toward the EU.
Imports from EU share	MEU	Trade-extension outcome	IMF Direction of Trade Statistics	Computed as 100 times imports from the EU divided by imports from the world.	Measures import composition toward the EU.

Table 4: Missing-data treatment in the reconstructed panel

Variable or issue	Method used	Notes
Polity score	Nearest-observation fill	Applied generally to fill missing institutional observations.
Agriculture	FAOSTAT supplement; nearest-observation fill	FAOSTAT was used to fill missing WDI agriculture values where possible. Remaining gaps were handled through the imputation plan.
Industry	Nearest-observation fill in data construction; excluded from final SCM	Although industry was included in the broader data construction process, it was dropped from final SCM because data quality remained problematic.
ODA	Country-specific and general nearest-observation fill	Several donor countries required endpoint filling. ODA construction remains a reconstruction choice because the original paper does not clearly document the denominator.
FDI	Zero-fill for selected early gaps; nearest-observation fill generally	Zero-filling was used for selected cases where missing early values were treated as approximately zero; other gaps were filled using nearest observations.
Labour	Country-specific and general nearest-observation fill	Used for countries with early, late, or broad labour-data gaps.
Government share	Nearest-observation fill	Constructed from PWT and filled where necessary.
Investment share	Nearest-observation fill	Constructed from PWT and filled where necessary.
GDP variables	Nearest-observation fill; selected linear interpolation	Several GDP alternatives were constructed and filled during data preparation, though the final outcome was WDI GDP per capita in current US dollars.
Donor eligibility	Donor screening before SCM	Donors entirely missing at least one active predictor over the pre-treatment period were excluded from the usable SCM donor pool.

A Issues in the Original Paper and Implications for Replication

A.1 Primary issues

A major obstacle in the replication was identifying the GDP variable used in the original paper. Although the paper describes the outcomes as real GDP per capita, the reported values could not be matched to any standard GDP series available in either WDI or Penn World Table. We tested the full set of WDI GDP per capita measures, PWT output- and expenditure-side GDP variables, and constructed alternatives combining WDI aggregate GDP measures with PWT population to generate per-capita series. None of these reproduced the GDP values reported in the original tables. This created a fundamental reproducibility problem, because GDP per capita is both the main outcome and a key lagged predictor in the SC specification. We therefore treated the GDP series as an empirical reconstruction choice, rather than a directly observed replication input, selecting WDI GDP per capita in current US dollars, because it provided the closest correspondence to the published synthetic control patterns.

This is important because synthetic control estimation requires a usable pre-treatment panel: missing values in the outcome or predictors can change which countries are eligible donors, alter predictor averages, and therefore affect the estimated synthetic counterfactual. In reconstructing the GDP analysis, missingness was substantial for several variables used or discussed in the original specification, especially sectoral predictors and some macro controls. The paper does not explain whether missing observations were dropped, interpolated, extrapolated, filled from alternative sources, or otherwise adjusted. We therefore had to make these choices explicitly. The final replication keeps both unimputed and imputed panels, fills selected gaps using documented

rules, supplements agriculture with an additional source where possible, and drops industry from the final SCM specification because retaining it created serious data-availability and quality problems.

These two primary issues have major implications for the methods and results of this analysis, and also have many possible approaches. As such, we did not expect to be able to exactly match the results of the paper, but rather aimed to produce a transparent reconstruction of the analysis, with clearly documented choices.

A.2 Secondary Issues

In addition to the missing methodology for missing data, as well as the unclear source reporting and unidentifiable GDP data, there are a number of smaller issues identified in the paper. Some of these issues have moderate implications for the results, while some should have no major impact. However, the number of directly observable mistakes in the paper substantially reduces confidence in the methods and underlying work that are not described or explicitly shown.

Inflation source

The original paper describes the inflation data coming from WDI, however the published table values match the Penn World Table deflator-based series. This is unexpected, given the paper’s text explicitly frames the inflation comparison as being calculated from WDI values. This issues raised questions about the sources of other variables, when replication with the listed sources was not possible.

Donor-country naming: Dominica vs Dominican Republic

The original paper’s donor-pool description refers to the “Dominican Rep.”, while the synthetic control weights table lists “Dominica”. There are different countries, with different ISO codes, economic countries, and data histories. In the replication, we chose to align with the donor countries listed in the table and use Dominica.

Donor-pool size

The paper states that each treated country is matched using 21 control countries, but the donor-weight table only lists 20 countries. This brings some uncertainty about the exact donor countries used in the published analysis.

DRC label in CFA-zone tables

One of the published predictor-balance tables labels a treated country as “DRC”. The Democratic Republic of Congo is not a CFA franc-zone country and is not part of WAEMU or CEMAC. This is likely a mislabelling of the Republic of Congo, but it is also perhaps an indication that the wrong country was used as a treated country in the analysis. The replication uses explicit ISO3 country codes throughout to avoid ambiguity.

Devaluation dummy

The paper reports a 1994 CFA devaluation dummy as part of the predictor balance, but the values do not appear consistent with a dummy variable averaged over the stated pre-treatment period. Additionally, Senegal is reported as zero despite being a CFA country in 1994. For this reason, the devaluation dummy is not retained as a final SCM predictor in the replication.

Repeated GDP values in selected years

The GDP values reported in Appendix 4 of the paper repeat exactly across different years for the same country. This is clearly a mistake in the table, but means that one of the two columns is missing. This error was made more challenging by the inability to find their reported numbers through any calculation method with their listed sources.

Investment variable definition

The paper’s variable appendix appears to describe investment as a household-consumption share. This is inconsistent with the usual interpretation of investment of gross fixed capital formation. Since investment is an important growth predictor, this creates uncertainty over the exact variable used in the analysis: whether an incorrect variable was used, or whether the correct variables was used but not correctly reported.

Sectoral variable sources

The paper lists agriculture and industry as coming from WDI and Penn World Table, but Penn World Table does not provide the same standard sectoral value-added shares used in WDI. This makes the source documentation

ambiguous. In the replication, agriculture is primarily taken from WDI and supplemented with FAOSTAT where needed, while industry is dropped because of missingness and data-quality problems.

OLS appendix observation counts

The paper's OLS appendix reports 751 observations for country-specific regressions over 1990-2019. For 13 countries over 29 years, we would expect 377 observations. It is unclear what the 751 observations are, or how this table was generated, but this value suggests some error in calculation or duplication.

FDI copy error in OLS table

The paper's OLS appendix has the reported FDI coefficient for Senegal as -43.7, vastly out of scale with the reported FDI coefficients for every other country (all between 15816 and 17245). Additionally, this value and its standard error match exactly the value for the ODA coefficient, suggesting there are errors in the table generation, possibly mistakenly copying the wrong value in.

Post-treatment RMSPE wording

The methodology section states that weights are chosen to minimise post-treatment RMSPE, but standard synthetic control methods chose weights using pre-treatment information only. Using post-treatment outcomes to choose weights would violate the logic of the method. This is likely a wording error, but would substantially change the method and results if it were the case.

Inflation table period label

The first column of the WAEMU inflation table appears to be incorrectly labelled. The table header indicates 1980-2021, while the other tables and surrounding text reference the pre-euro and post-euro period. These numbers do not match those identified by our replication for either the pre-euro, or total periods. Our ability to match the other inflation table values, except for the WAEMU section, paired with the incorrect label, suggests there may have been a mistake in the table creation or value calculations.

Zero inflation value for Sao Tome and Principe

The non-CFA inflation table reports Sao Tome and Principe as having exactly zero average inflation over the pre-period. This reflects missing data treatment rather than an actual inflation average, and the replication exercise finds a positive value here (31.7%) along with alignment with the rest of the table. This is one instance where missing value treatment is not explained but may have impacts on the results of the paper.

Synthetic control weights reported as percentage points

The paper describes the synthetic-control weights as being expressed in percentage points, but the values in the table are decimal weights. This is just a terminology issue, but matters for the interpretation of the table.